Skanda Bhairava

CV

About me

Computer Science student interested in distributed systems, and high performance computing. I enjoy learning by building tools and working on projects.

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Bangalore, India

Languages

Python • Go • C++ • CUDA • Rust • JS/TS

Tools

PyTorch • RayCore • Svelte • Gin • FastAPI • PyGame

Interests

Distributed Systems • High Performance Computing(HPC) • Simulations • Graphics • Research

Experience

June 2025 — **Kaynes Technology**
Software Engineer Intern · Mysuru, India
Designed and built internal tools for RFID (Radio Frequency Identification) reader configuration, DMI (Driver Machine Interface) testing and UI debugging. These tools were built for The Kavach project implimented in the Indian Railways.

Education

2023–Current — **Bachelors** ·
Information Technology (Honours)
Manipal Institute of Technology Bangalore
CGPA: 9.26/10

Honours and Awards

2026 — **Best Research Poster Award, Turinger Symposium**
Manipal Institute of Technology · Bangalore
Received the Best Poster Award for research, at the Turinger Symposium, Manipal Institute of Technology Bangalore, February 2026.

2026 — **Hackathon Finalist**
HCL GUVI · AI Impact Summit 2026, Delhi
Our team was placed in the top 2% in the HCL GUVI hackathon for developing a Honeypot to extract details from a Scammer in a textual environment.

2024, 2025 — **Manipal Achievers Scholarship (Dean's List)**
Manipal Academy of Higher Education · Bangalore
Received the Manipal Achievers Scholarship in 2024 and 2025 for maintaining a CGPA of 9.26 out of 10.

Publications

[First Author] Skanda M Bhairava, Gururaj H L. "ADaRES: Adaptive Data Reallocation and Epoch Scheduling in Federated Learning Under Network and Resource Constraints". COMSNETS 2026
doi.org/10.1109/COMSNETS67989.2026.11418093

Projects (Top 6)

HierarchicalFLSim — Python • RayCore • Systems
Built a configurable Hierarchical Federated Learning simulator used to experiment with flexible topology, Non-IID distributions, client dropout and custom aggregation algorithms. The simulator also helps validate Cost model for hierarchical training.

Leaf Disease Detection — Python • Pytorch • ML
Developed a lightweight CNN that performed better than baseline architectures (AlexNet, DeepNet) on a leaf disease dataset in important metrics.

Raytracer — C++ • Multithreading • Graphics
Implemented a 3D Ray Tracing Engine from scratch in C++ using hand-derived math, multithreading and a custom Pixel Representation Format.

GravBall — Python • PyGame • Simulation
Designed and Optimized a Physics based Python game to run at 120 fps by redesigning collision and rendering loops.

Obbattu — Rust, TS • Svelte, OracleVPS • Full Stack
Built a full stack crossword generator with daily puzzle regeneration using a backtracking solver, with an inbuilt recovery mechanism after host downtimes. Deployed on VPS with Rust backend and Svelte frontend.

DragonCommissions — Python • Automation • Management
Co-founded a plugin development hub that built 100+ custom plugins for gaming creators. Coordinated a team of around 20 developers, and handled client requirements for around 10+ creators. I also built internal automation tools and pipelines that made development workflows efficient.